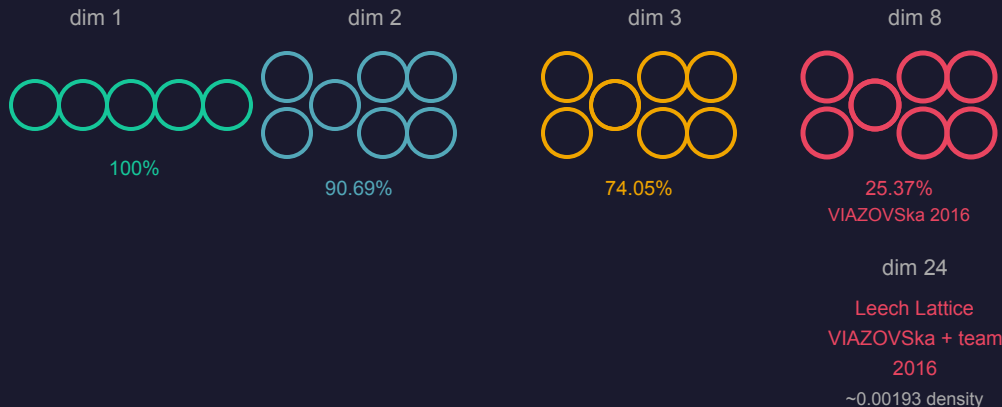


Sphere Packing: From Kepler to Viazovska



The Cohn--Kenyon Method

Construct an auxiliary function f with:

$$f(x) \leq 0 \text{ for } \|x\| \geq 1 \text{ AND } \hat{f}(x_i) \leq 0 \text{ for } 0 < \|x_i\| \leq 1$$

Viazovska: f is built from modular forms!

This exact construction works only in dim 8 and 24